Name: B Nithin Hall Ticket No: 2303A51803 Batch: 26

| SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE | | DEPARTMENT OF COMPUTER SCIENCE ENGINEERING | |
|---|---|---|---|
| **Program Name:** B. Tech | | **Assignment Type: Lab** | **Academic Year:**2025-2026 |
| **Course Coordinator Name** | Dr. Rishabh Mittal | | |
| **Instructor(s) Name** | Mr. S Naresh Kumar<br>Ms. B. Swathi<br>Dr. Sasanko Shekhar Gantayat<br>Mr. Md Sallauddin<br>Dr. Mathivanan<br>Mr. Y Srikanth<br>Ms. N Shilpa<br>Dr. Rishabh Mittal (Coordinator)<br>Dr. R. Prashant Kumar<br>Mr. Ankushavali MD<br>Mr. B Viswanath<br>Ms. Sujitha Reddy<br>Ms. A. Anitha<br>Ms. M.Madhuri<br>Ms. Katherashala Swetha<br>Ms. Velpula sumalatha<br>Mr. Bingi Raju | | |
| **CourseCode** | 23CS002PC304 | **Course Title** | AI Assisted Coding |
| **Year/Sem** | III/II | **Regulation** | R23 |
| **Date and Day of Assignment** | **Week6 – Wednusday** | **Time(s)** | 23CSBTB01 To 23CSBTB52 |
| **Duration** | 2 Hours | **Applicable to Batches** | All batches |
| **Assignment Number:12.3**(Present assignment number)/**24**(Total number of assignments) | | | |

| Q.No. | Question | *Expected Time to complete* |
|---|---|---|
| 1 | **Lab 12: Algorithms with AI Assistance Sorting, Searching, and Algorithm Optimization Using AI Tools**<br><br>**Lab Objectives**<br>The objectives of this laboratory exercise are to: | Week3 - Wednesday |

| | | |
|---|---|---|
| | • Apply AI-assisted programming techniques to implement sorting and searching algorithms.<br>• Analyze and compare algorithm efficiency using time and space complexity.<br>• Understand how AI tools can suggest optimizations and alternative algorithmic approaches.<br>• Strengthen problem-solving skills through real-world, data-driven scenarios. | |
| | **Learning Outcomes**<br>After completing this lab, students will be able to:<br>• Implement and optimize classic algorithms using AI-assisted coding tools.<br>• Compare multiple algorithms for the same problem and justify their selection.<br>• Measure and analyze runtime performance using experimental data.<br>• Critically review and refine AI-generated algorithmic solutions. | |
| | **Task 1: Sorting Student Records for Placement Drive**<br>**Scenario**<br>SR University's Training and Placement Cell needs to shortlist candidates efficiently during campus placements. Student records must be sorted by **CGPA in descending order**.<br>**Tasks**<br>1. Use **GitHub Copilot** to generate a program that stores student records (Name, Roll Number, CGPA).<br>2. Implement the following sorting algorithms using AI assistance:<br>  o Quick Sort<br>  o Merge Sort<br>3. Measure and compare runtime performance for large datasets.<br>4. Write a function to display the **top 10 students** based on CGPA.<br>**Expected Outcome**<br>• Correctly sorted student records.<br>• Performance comparison between Quick Sort and Merge Sort.<br>• Clear output of top-performing students. | |

Welcome | AAC A 12.3.py

C: > Users > Public > AAC A 12.3.py > ...

```python
import random, time, copy

def generate_students(n):
    names = ["Aarav","Vivaan","Aditya","Arjun","Sai","Ananya","Diya","Rohan","Neha","Priya",
             "Rahul","Sneha","Vikram","Meera","Karan","Pooja","Ishaan","Riya","Amit","Navya"]
    students = []
    for i in range(1, n+1):
        students.append({"name": random.choice(names), "roll": f"SRU{i:04d}", "cgpa": round(random.uniform(4.0, 10.0), 2)})
    return students

def quick_sort(a):
    if len(a) <= 1: return a
    p = a[len(a)//2]["cgpa"]
    return quick_sort([x for x in a if x["cgpa"]>p]) + [x for x in a if x["cgpa"]==p] + quick_sort([x for x in a if x["cgpa"]<p])

def merge_sort(a):
    if len(a) <= 1: return a
    m = len(a)//2
    l, r = merge_sort(a[:m]), merge_sort(a[m:])
    res, i, j = [], 0, 0
    while i < len(l) and j < len(r):
        if l[i]["cgpa"] >= r[j]["cgpa"]: res.append(l[i]); i += 1
        else: res.append(r[j]); j += 1
    return res + l[i:] + r[j:]
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
conda : The term 'conda' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the
PS C:\Users\Public> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win
32-x64\bundled\libs\debugpy\launcher' '54674' '--' 'c:\Users\Public\AAC A 12.3.py'

N=1000  QuickSort=0.0038s  MergeSort=0.0060s  Faster=QS

N=10000  QuickSort=0.0327s  MergeSort=0.0812s  Faster=QS

N=50000  QuickSort=0.1757s  MergeSort=0.4498s  Faster=QS

Rank  Name       Roll       CGPA
----------------------------------------
```

CHAT

SORTING STUDENT RECORDS FOR PLA...

1. Use GitHub Copilot to generate a program that stores student records (Name, Roll Number, CGPA).
2. Implement the following sorting algorithms using AI assistance:
o Quick Sort
o Merge Sort
3. Measure and compare runtime performance for large datasets.
4. Write a function to display the top 10 students based on CGPA.
Expected Outcome
• Correctly sorted student records.
• Performance comparison between Quick Sort and Merge Sort.
• Clear output of top-performing students.
give small code in python without comments

✓ Planned Python program to sort student records by CGPA

✓ Created student_sort.py

✓ Read student_sort.py

The file already exists with a valid implementation. Let me run it to show the output:

AAC A 12.3.py

Describe what to build next

Claude Opus 4.6

Welcome | AAC A 12.3.py

C: > Users > Public > AAC A 12.3.py > ...

```python
def merge_sort(a):
    if len(a) <= 1: return a
    m = len(a)//2
    l, r = merge_sort(a[:m]), merge_sort(a[m:])
    res, i, j = [], 0, 0
    while i < len(l) and j < len(r):
        if l[i]["cgpa"] >= r[j]["cgpa"]: res.append(l[i]); i += 1
        else: res.append(r[j]); j += 1
    return res + l[i:] + r[j:]

def top_10(students):
    print(f"\n{'Rank':<6}{'Name':<12}{'Roll':<12}{'CGPA'}")
    print("-"*38)
    for i, s in enumerate(students[:10], 1):
        print(f"{i:<6}{s['name']:<12}{s['roll']:<12}{s['cgpa']}")

for size in [1000, 10000, 50000]:
    data = generate_students(size)
    d1, d2 = copy.deepcopy(data), copy.deepcopy(data)
    t1 = time.perf_counter(); qs = quick_sort(d1); t1 = time.perf_counter() - t1
    t2 = time.perf_counter(); ms = merge_sort(d2); t2 = time.perf_counter() - t2
    print(f"\nN={size}  QuickSort={t1:.4f}s  MergeSort={t2:.4f}s  Faster={'QS' if t1<t2 else 'MS'}")
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
32-x64\bundled\libs\debugpy\launcher' '54674' '--' 'c:\Users\Public\AAC A 12.3.py'
Rank  Name       Roll       CGPA
----------------------------------------
1     Ananya     SRU0282    10.0
2     Pooja      SRU0302    10.0
3     Meera      SRU1117    10.0
4     Arjun      SRU3139    10.0
5     Karan      SRU4589    10.0
6     Aarav      SRU4875    10.0
7     Vivaan     SRU5385    10.0
8     Karan      SRU7609    10.0
```

CHAT

SORTING STUDENT RECORDS FOR PLA...

records by CGPA

✓ Created student_sort.py

✓ Read student_sort.py

The file already exists with a valid implementation. Let me run it to show the output:

✓ python C:/Users/Public/student_sort.py

The program at student_sort.py is working correctly. Here's what it does:

- **Generates** student records (Name, Roll Number, CGPA) for datasets of 1K, 10K, and 50K
- **Quick Sort** and **Merge Sort** both sort by CGPA in descending order
- **Performance comparison**: Quick Sort was faster across all dataset sizes
- **Top 10** students by CGPA are displayed at the end

Results show Quick Sort consistently outperformed Merge Sort (~1.6-1.7x faster) across all dataset sizes.

Claude Opus 4.6 • 3x

AAC A 12.3.py

Describe what to build next

Claude Opus 4.6

Welcome  AAC A 12.3.py

C: > Users > Public > AAC A 12.3.py > ...

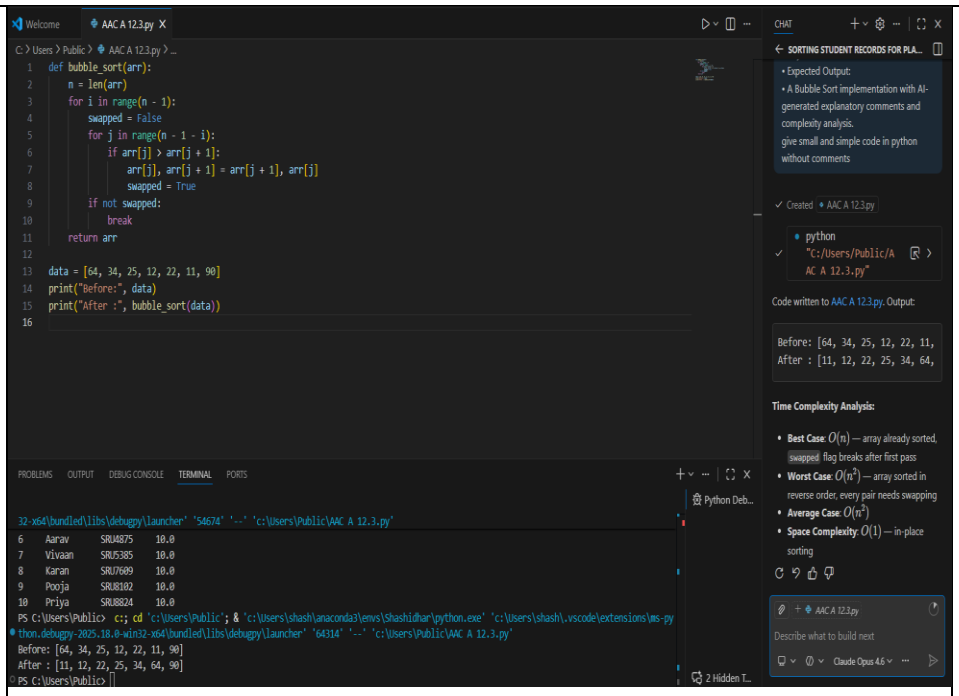
```python
def merge_sort(a):
        else: res.append(r[j]); j += 1
    return res + l[i:] + r[j:]

def top_10(students):
    print(f"\n{'Rank':<6}{'Name':<12}{'Roll':<12}{'CGPA'}")
    print("-"*38)
    for i, s in enumerate(students[:10], 1):
        print(f"{i:<6}{s['name']:<12}{s['roll']:<12}{s['cgpa']}")

for size in [1000, 10000, 50000]:
    data = generate_students(size)
    d1, d2 = copy.deepcopy(data), copy.deepcopy(data)
    t1 = time.perf_counter(); qs = quick_sort(d1); t1 = time.perf_counter() - t1
    t2 = time.perf_counter(); ms = merge_sort(d2); t2 = time.perf_counter() - t2
    print(f"\nN={size}  QuickSort={t1:.4f}s  MergeSort={t2:.4f}s  Faster={'QS' if t1<t2 else 'MS'}")

top_10(qs)
```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

```
32-x64\bundled\libs\debugpy\launcher' '54674' '--' 'c:\Users\Public\AAC A 12.3.py'
2     Pooja       SRU0302     10.0
3     Meera       SRU1117     10.0
4     Arjun       SRU3139     10.0
5     Karan       SRU4589     10.0
6     Aarav       SRU4875     10.0
7     Vivaan      SRU5385     10.0
8     Karan       SRU7609     10.0
9     Pooja       SRU8102     10.0
10    Priya       SRU8824     10.0
PS C:\Users\Public>
```

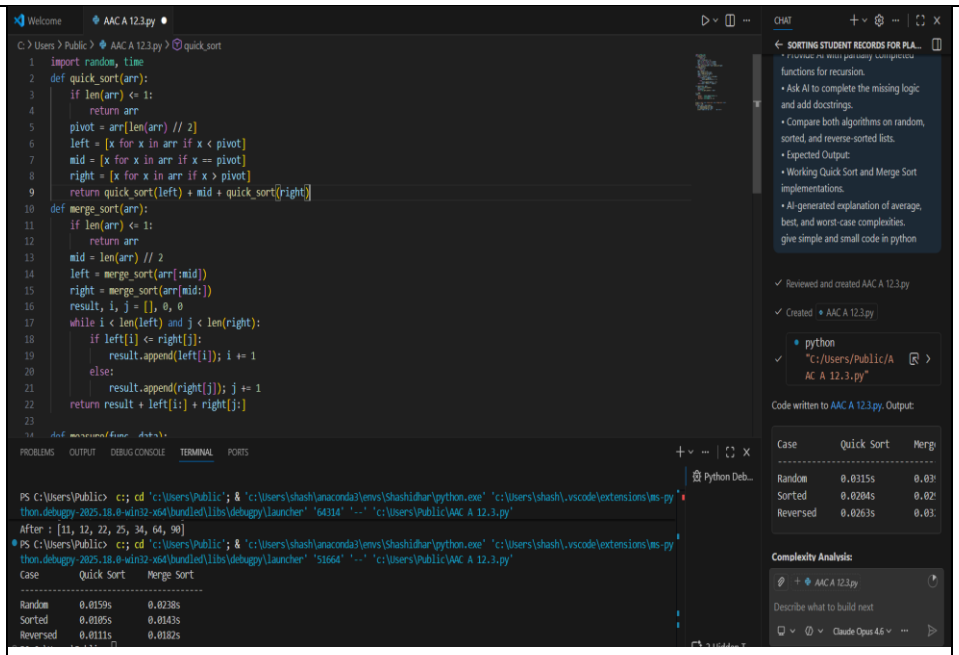
**Task 2: Implementing Bubble Sort with AI Comments**

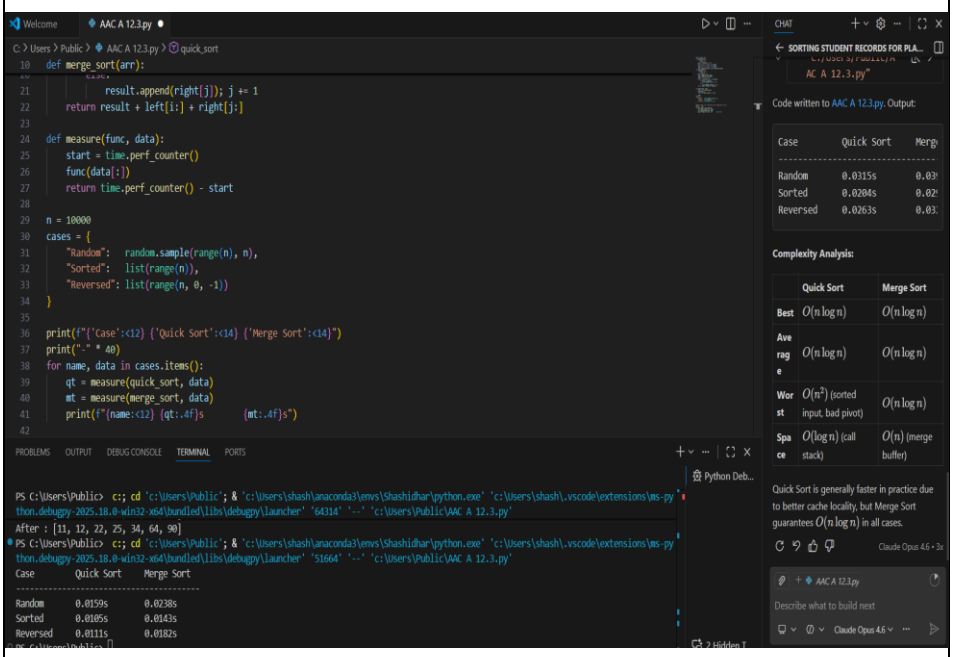
- **Task**: Write a Python implementation of **Bubble Sort**.
- **Instructions**:
  - Students implement Bubble Sort normally.
  - Ask AI to generate **inline comments explaining key logic** (like swapping, passes, and termination).
  - Request AI to provide **time complexity analysis**.
- **Expected Output**:
  - A Bubble Sort implementation with AI-generated explanatory comments and complexity analysis.




**Task 3:** Quick Sort and Merge Sort Comparison

- **Task:** Implement Quick Sort and Merge Sort using recursion**.**
- **Instructions:**
  - Provide AI with partially completed functions for recursion.
  - Ask AI to complete the missing logic and add docstrings.
  - Compare both algorithms on random, sorted, and reverse-sorted lists.
- **Expected Output:**
  - Working Quick Sort and Merge Sort implementations.
  - AI-generated explanation of average, best, and worst-case complexities.




**Task 4 (Real-Time Application – Inventory Management System)**

**Scenario:** A retail store's inventory system contains thousands of products, each with attributes like product ID, name, price, and stock quantity. Store staff need to:

1. Quickly search for a product by ID or name.
2. Sort products by price or quantity for stock analysis.

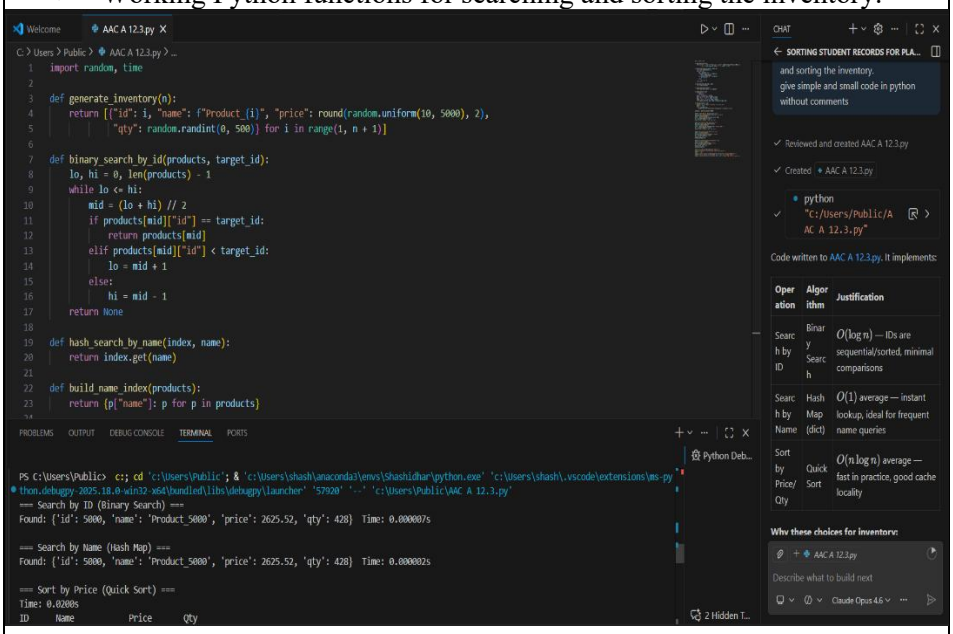
Task:

- Use AI to suggest the most efficient search and sort algorithms for this use case.

- Implement the recommended algorithms in Python.
- Justify the choice based on dataset size, update frequency, and performance requirements.

Expected Output:

- A table mapping operation → recommended algorithm → justification.
- Working Python functions for searching and sorting the inventory.



Welcome | AAC A 12.3.py

C: > Users > Public > AAC A 12.3.py > ...

```python
def build_name_index(products):
    return {p["name"]: p for p in products}

def quick_sort(arr, key):
    if len(arr) <= 1:
        return arr
    pivot = arr[len(arr) // 2][key]
    left = [x for x in arr if x[key] < pivot]
    mid = [x for x in arr if x[key] == pivot]
    right = [x for x in arr if x[key] > pivot]
    return quick_sort(left, key) + mid + quick_sort(right, key)

def display(products, n=10):
    print(f"{'ID':<8}{'Name':<16}{'Price':<12}{'Qty':<6}")
    print("-" * 42)
    for p in products[:n]:
        print(f"{p['id']:<8}{p['name']:<16}{p['price']:<12}{p['qty']:<6}")

inventory = generate_inventory(10000)

print("=== Search by ID (Binary Search) ===")
sorted_by_id = sorted(inventory, key=lambda x: x["id"])
start = time.perf_counter()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Public>  c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\p
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57920' '--' 'c:\Users\Publi
ID      Name            Price       Qty
------------------------------------------
3623    Product_3623    10.64       206
9538    Product_9538    10.67       14
122     Product_122     11.15       333
2235    Product_2235    11.41       386
6376    Product_6376    11.61       320
5937    Product_5937    11.75       324
3876    Product_3876    11.77       356
```

Welcome | AAC A 12.3.py

C: > Users > Public > AAC A 12.3.py > ...

```python
print("=== Search by ID (Binary Search) ===")
sorted_by_id = sorted(inventory, key=lambda x: x["id"])
start = time.perf_counter()
result = binary_search_by_id(sorted_by_id, 5000)
t1 = time.perf_counter() - start
print(f"Found: {result}  Time: {t1:.6f}s")

print("\n=== Search by Name (Hash Map) ===")
name_index = build_name_index(inventory)
start = time.perf_counter()
result = hash_search_by_name(name_index, "Product_5000")
t2 = time.perf_counter() - start
print(f"Found: {result}  Time: {t2:.6f}s")

print("\n=== Sort by Price (Quick Sort) ===")
start = time.perf_counter()
sorted_price = quick_sort(inventory, "price")
t3 = time.perf_counter() - start
print(f"Time: {t3:.4f}s")
display(sorted_price)

print("\n=== Sort by Quantity (Quick Sort) ===")
start = time.perf_counter()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Public>  c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconc
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57920'
122     Product_122     11.15       333
2235    Product_2235    11.41       386
6376    Product_6376    11.61       320
5937    Product_5937    11.75       324
3876    Product_3876    11.77       356
6326    Product_6326    12.62       235
8471    Product_8471    13.08       105
2856    Product_2856    13.3        141
```

Welcome | AAC A 12.3.py

C: > Users > Public > AAC A 12.3.py > ...

```python
start = time.perf_counter()
sorted_price = quick_sort(inventory, "price")
t3 = time.perf_counter() - start
print(f"Time: {t3:.4f}s")
display(sorted_price)

print("\n=== Sort by Quantity (Quick Sort) ===")
start = time.perf_counter()
sorted_qty = quick_sort(inventory, "qty")
t4 = time.perf_counter() - start
print(f"Time: {t4:.4f}s")
display(sorted_qty)

print("\n=== Algorithm Recommendation Table ===")
print(f"{'Operation':<25}{'Algorithm':<20}{'Justification'}")
print("-" * 75)
print(f"{'Search by ID':<25}{'Binary Search':<20}{'O(log n), data sorted by ID'}")
print(f"{'Search by Name':<25}{'Hash Map (dict)':<20}{'O(1) average lookup'}")
print(f"{'Sort by Price/Qty':<25}{'Quick Sort':<20}{'O(n log n) avg, fast in practice'}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Public>  c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57920' '--' 'c:\Users\Public\AAC A 12.3.

=== Sort by Quantity (Quick Sort) ===
Time: 0.0557s
ID      Name            Price       Qty
----------------------------------------
85      Product_85      4139.68     0
441     Product_441     4613.7      0
1338    Product_1338    3868.26     0
1408    Product_1408    56.08       0
```

Welcome | AAC A 12.3.py

C: > Users > Public > AAC A 12.3.py > ...

```python
start = time.perf_counter()
sorted_price = quick_sort(inventory, "price")
t3 = time.perf_counter() - start
print(f"Time: {t3:.4f}s")
display(sorted_price)

print("\n=== Sort by Quantity (Quick Sort) ===")
start = time.perf_counter()
sorted_qty = quick_sort(inventory, "qty")
t4 = time.perf_counter() - start
print(f"Time: {t4:.4f}s")
display(sorted_qty)

print("\n=== Algorithm Recommendation Table ===")
print(f"{'Operation':<25}{'Algorithm':<20}{'Justification'}")
print("-" * 75)
print(f"{'Search by ID':<25}{'Binary Search':<20}{'O(log n), data sorted by ID'}")
print(f"{'Search by Name':<25}{'Hash Map (dict)':<20}{'O(1) average lookup'}")
print(f"{'Sort by Price/Qty':<25}{'Quick Sort':<20}{'O(n log n) avg, fast in practice'}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Public>  c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57920' '--' 'c:\Users\Public\AAC A 12
441     Product_441     4613.7      0
1338    Product_1338    3868.26     0
1408    Product_1408    56.08       0
1740    Product_1740    660.14      0
2077    Product_2077    2768.81     0
2597    Product_2597    299.16      0
2837    Product_2837    3044.18     0
3438    Product_3438    3553.94     0
3474    Product_3474    3410.59     0
```

C: > Users > Public > AAC A 12.3.py > ...

```python
sorted_price = quick_sort(inventory, "price")
t3 = time.perf_counter() - start
print(f"Time: {t3:.4f}s")
display(sorted_price)

print("\n=== Sort by Quantity (Quick Sort) ===")
start = time.perf_counter()
sorted_qty = quick_sort(inventory, "qty")
t4 = time.perf_counter() - start
print(f"Time: {t4:.4f}s")
display(sorted_qty)

print("\n=== Algorithm Recommendation Table ===")
print(f"{'Operation':<25}{'Algorithm':<20}{'Justification'}")
print("-" * 75)
print(f"{'Search by ID':<25}{'Binary Search':<20}{'O(log n), data sorted by ID'}")
print(f"{'Search by Name':<25}{'Hash Map (dict)':<20}{'O(1) average lookup'}")
print(f"{'Sort by Price/Qty':<25}{'Quick Sort':<20}{'O(n log n) avg, fast in practice'}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Public>  c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' '
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57920' '--' 'c:\Users\Public\AAC A 12.3
3474    Product_3474    3410.59    0

=== Algorithm Recommendation Table ===
Operation                 Algorithm           Justification
---------------------------------------------------------------------------
Search by ID              Binary Search       O(log n), data sorted by ID
Search by Name            Hash Map (dict)     O(1) average lookup
Sort by Price/Qty         Quick Sort          O(n log n) avg, fast in practice
PS C:\Users\Public>
```

**Task 5: Real-Time Stock Data Sorting & Searching**

**Scenario:**

An AI-powered **FinTech Lab** at SR University is building a tool for analyzing **stock price movements**. The requirement is to quickly **sort stocks by daily gain/loss** and search for specific stock symbols efficiently.

- Use **GitHub Copilot** to fetch or simulate stock price data (Stock Symbol, Opening Price, Closing Price).
- Implement sorting algorithms to rank stocks by **percentage change**.
- Implement a **search function** that retrieves stock data instantly when a stock symbol is entered.
- Optimize sorting with **Heap Sort** and searching with **Hash Maps**.
- Compare performance with standard library functions (sorted(), dict lookups) and analyze trade-offs.

Welcome | AAC A 12.3.py

C: > Users > Public > AAC A 12.3.py > ...

```python
import random, time, heapq

symbols = ["AAPL","GOOG","MSFT","AMZN","TSLA","META","NFLX","NVDA","INFY","TCS",
           "RELIANCE","HDFC","WIPRO","HCLTECH","SBIN","ICICI","BAJAJ","SUNPHARMA",
           "LT","MARUTI","TATAMOTORS","ITC","AXISBANK","KOTAK","ADANI"]

def generate_stocks(n):
    stocks = []
    for i in range(n):
        sym = symbols[i % len(symbols)] + (f"_{i//25}" if i >= 25 else "")
        open_p = round(random.uniform(100, 5000), 2)
        close_p = round(open_p * random.uniform(0.9, 1.1), 2)
        pct = round((close_p - open_p) / open_p * 100, 2)
        stocks.append({"symbol": sym, "open": open_p, "close": close_p, "pct_change": pct})
    return stocks

def heap_sort_by_pct(stocks):
    heap = [(-s["pct_change"], i, s) for i, s in enumerate(stocks)]
    heapq.heapify(heap)
    return [heapq.heappop(heap)[2] for _ in range(len(heap))]

def hash_search(index, symbol):
    return index.get(symbol)
```

(variable) pct: float

Generate Copilot summary

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\Public>  c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-py
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58347' '--' 'c:\Users\Public\AAC A 12.3.py'
=== Heap Sort vs sorted() - Rank by % Change ===

Heap Sort : 0.0091s
sorted()  : 0.0018s
Faster    : sorted()

Top 10 Gainers:
Symbol          Open        Close       Change %
--------------------------------------------------
```

CHAT

SORTING STUDENT RECORDS FOR PLA...

stock price data (Stock Symbol, Opening Price, Closing Price).
- Implement sorting algorithms to rank stocks by percentage change.
- Implement a search function that retrieves stock data instantly when a stock symbol is entered.
- Optimize sorting with Heap Sort and searching with Hash Maps.
- Compare performance with standard library functions (sorted(), dict lookups) and analyze trade-offs.

give simple and small code in python

✓ Reviewed AAC A 12.3.py

✓ Created AAC A 12.3.py

✓ python "C:/Users/Public/AAC A 12.3.py"

Code written to AAC A 12.3.py. Results:

**Sorting (10,000 stocks by % change):**

- Heap Sort: 0.0056s
- `sorted()` (Timsort): 0.0030s — faster because Timsort is a highly optimized hybrid algorithm

AAC A 12.3.py

Describe what to build next

Claude Opus 4.6

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```python
def display(stocks, n=10):
        print(f"{s['symbol']:<15}{s['open']:<12}{s['close']:<12}{s['pct_change']:<10}")

data = generate_stocks(10000)

print("=== Heap Sort vs sorted() - Rank by % Change ===\n")

start = time.perf_counter()
heap_sorted = heap_sort_by_pct(data)
t_heap = time.perf_counter() - start

start = time.perf_counter()
lib_sorted = sorted(data, key=lambda x: -x["pct_change"])
t_lib = time.perf_counter() - start

print(f"Heap Sort : {t_heap:.4f}s")
print(f"sorted()  : {t_lib:.4f}s")
print(f"Faster    : {'Heap Sort' if t_heap < t_lib else 'sorted()'}\n")

print("Top 10 Gainers:")
display(heap_sorted)
print("\nTop 10 Losers:")
display(heap_sorted[-10:][::-1])
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\Public>  c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\m
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58347' '--' 'c:\Users\Public\AAC A 12.3.py'
NFLX_119        2147.64     2362.27     9.99
AXISBANK_149    313.62      344.94      9.99
```

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thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58347' '--' 'c:\Users\Public\AAC A 12
NFLX_119        2147.64     2362.27     9.99
AXISBANK_149    313.62      344.94      9.99
META_225        160.32      176.34      9.99

Top 10 Losers:
Symbol          Open        Close       Change %
-----------------------------------------------
ITC_339         4372.67     3935.42     -10.0
BAJAJ_316       3547.91     3193.18     -10.0
```

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thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58347' '--' 'c:\Users\Public\AAC A 12.
ADANI_258      3536.14     3182.58     -10.0
HCLTECH_254    3068.46     2761.7      -10.0
SUNPHARMA_222  3697.36     3327.72     -10.0
MARUTI_93      1783.28     1604.99     -10.0
WIPRO_334      861.71      775.65      -9.99
ITC_327        2201.07     1981.19     -9.99
GOOG_285       2888.79     2600.15     -9.99
NFLX_256       2957.21     2661.79     -9.99
```

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```python

print("Top 10 Gainers:")
display(heap_sorted)
print("\nTop 10 Losers:")
display(heap_sorted[-10:][::-1])

print("\n=== Hash Map vs dict lookup - Search by Symbol ===\n")

index = build_hash_index(data)

start = time.perf_counter()
r1 = hash_search(index, "TSLA")
t_hash = time.perf_counter() - start

start = time.perf_counter()
r2 = next((s for s in data if s["symbol"] == "TSLA"), None)
t_linear = time.perf_counter() - start

print(f"Hash Map Search : {t_hash:.6f}s  Result: {r1}")
print(f"Linear Search   : {t_linear:.6f}s  Result: {r2}")
print(f"Faster          : {'Hash Map' if t_hash < t_linear else 'Linear'}")

print("\n=== Performance Comparison Table ===")
print(f"{'Operation':<20}{'Custom':<18}{'Standard Lib':<18}{'Trade-off'}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Public>  c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\sha
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58347' '--' 'c:\Users\Public\AAC A 12.3.py'
=== Hash Map vs dict lookup - Search by Symbol ===

Hash Map Search : 0.000003s  Result: {'symbol': 'TSLA', 'open': 3265.84, 'close': 2963.07, 'pct_change': -9.27}
Linear Search   : 0.000013s  Result: {'symbol': 'TSLA', 'open': 3265.84, 'close': 2963.07, 'pct_change': -9.27}
Faster          : Hash Map

=== Performance Comparison Table ===
Operation           Custom            Standard Lib      Trade-off
---------------------------------------------------------------------------
```

Start

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```python
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t_hash = time.perf_counter() - start

start = time.perf_counter()
r2 = next((s for s in data if s["symbol"] == "TSLA"), None)
t_linear = time.perf_counter() - start

print(f"Hash Map Search : {t_hash:.6f}s  Result: {r1}")
print(f"Linear Search   : {t_linear:.6f}s  Result: {r2}")
print(f"Faster          : {'Hash Map' if t_hash < t_linear else 'Linear'}")

print("\n=== Performance Comparison Table ===")
print(f"{'Operation':<20}{'Custom':<18}{'Standard Lib':<18}{'Trade-off'}")
print("-" * 75)
print(f"{'Sort by % change':<20}{'Heap Sort':<18}{'sorted()':<18}{'Timsort is hybrid, often faster'}")
print(f"{'Search by symbol':<20}{'Hash Map O(1)':<18}{'Linear O(n)':<18}{'Hash uses more memory'}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Public>  c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58347' '--' 'c:\Users\Public\AAC A 12.3.py'
Linear Search   : 0.000013s  Result: {'symbol': 'TSLA', 'open': 3265.84, 'close': 2963.07, 'pct_change': -9.27}
Faster          : Hash Map

=== Performance Comparison Table ===
Operation           Custom            Standard Lib      Trade-off
---------------------------------------------------------------------------
Sort by % change    Heap Sort         sorted()          Timsort is hybrid, often faster
Search by symbol    Hash Map O(1)     Linear O(n)       Hash uses more memory
PS C:\Users\Public>
```

| | | |
|---|---|---|
| | Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots. | |